



**Portolan map**  
Date unknown

From the 13th century on, sailors relied on portolan charts—maps showing compass bearings—to guide them between ports. This early chart of the Mediterranean depicts the navigational lines between hundreds of ports.

line indicating  
compass bearing



**Astrolabe**

Late 15th century

Developed over 2,000 years ago to sight stars and make astronomical calculations, astrolabes were later simplified to find latitude at sea by measuring the height of the Sun and stars.

# NAVIGATION TOOLS

IMPROVEMENTS IN INSTRUMENT DESIGN HAVE MADE NAVIGATION INCREASINGLY PRECISE

**Ancient navigators relied on the position in the sky of the Sun and the stars to determine their location and chart their course. Later, a compass and an accurate timepiece could be used to work out direction and location.**

**For much of history, sailors found** their latitude with tools such as sextants, astrolabes, and quadrants that indicated the angle of the Sun and stars above the horizon. From about a thousand years ago, compasses gave them a direction to sail in—a bearing. And from the 1700s, chronometers finally enabled them to work out their longitude. For most modern navigators, these instruments have been replaced by satellite systems.



**Navigator's compass**

c.1860

From the 13th century onward, navigators used a magnetic compass with a wire lozenge or metal needle, mounted to swing freely, to find north.



**Lodestone**  
c.1550–1600

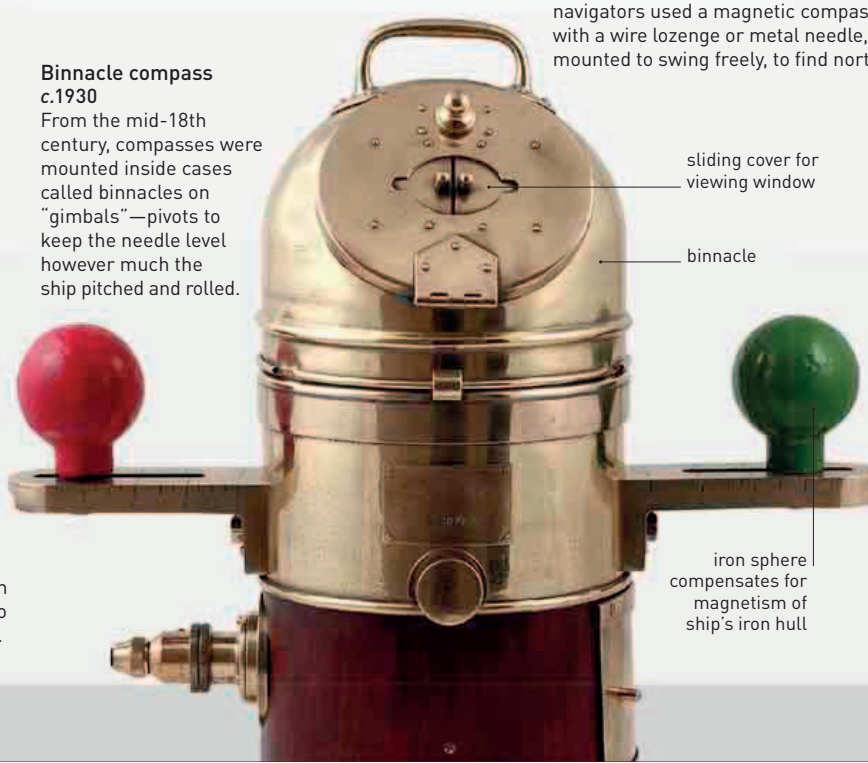
Chinese sailors used swinging lodestones—magnetic stones that turn to align with Earth's magnetic field—to gauge direction in overcast conditions.

magnetic  
mineral

**Binnacle compass**

c.1930

From the mid-18th century, compasses were mounted inside cases called binnacles on "gimbals"—pivots to keep the needle level however much the ship pitched and rolled.



sliding cover for  
viewing window

binnacle

iron sphere  
compensates for  
magnetism of  
ship's iron hull



**Marine chronometer**

c.1893

High-precision clocks, chronometers provided the accurate timekeeping necessary to keep track of longitude (distance east or west) on a long voyage.